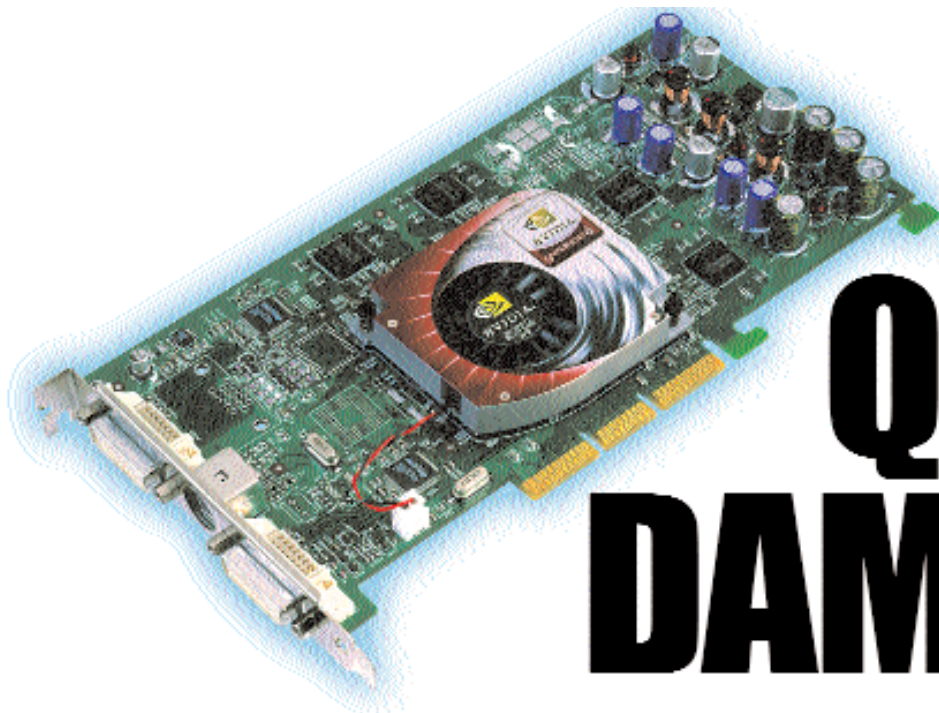




QUAD DAMAGE

With the introduction of its fourth generation of GeForce graphics cards, nVidia is all set to blow away the competition. Again

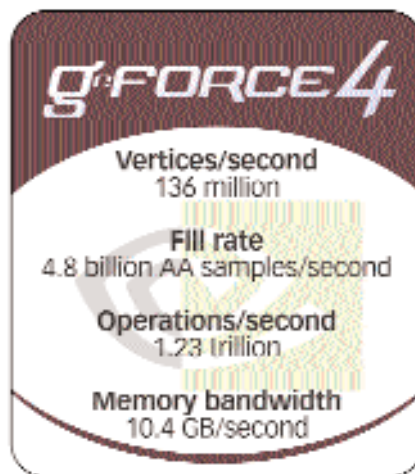
Everybody enjoys playing games on the PC. Well, almost everybody. And games on the PC have come a long way from the days when *Space Invaders* and *Pacman* ruled the roost. Today games such as *Max Payne* and *Ballistics* can entrance gamers largely due to improvements in graphics hardware over the past few years. But hold your breath. What's coming next is worlds apart.

The GeForce series of graphics cards from nVidia has taken several steps towards the holy grail of game developers—Total Reality. It gives them the ability to breathe life into their work, making it so real that suspension of disbelief in their handcrafted virtual worlds will be an easy matter. With the GeForce4, they promise to let you see the reflection of clouds in the ripples of a pond, and leaves actually fluttering under the gentle caress of an unseen breeze. This will be a world where you would see every minute detail, down to the twinkle in Duke's eye. God bless their souls.

Poetry on a PCB

One of the remarkable aspects about the GeForce series of cards is the fact that it utilises the state-of-the-art in manufacturing processes. The GeForce4 (code-named NV25) is made using 63 million transistors and uses a 0.15-micron

manufacturing process. Even Intel's Pentium 4, which has a relatively unimpressive 55 million transistors, was until recently fabricated using older 0.18-micron technology. The way the card's PCB is designed requires the use of a com-



plex eight-layer manufacturing process, which disqualifies the smaller video card manufacturing outfits from offering the top of the line GeForce4s.

In addition to this, nVidia has introduced a couple of innovations in the GeForce4, which are quite elegant from a technological standpoint. The most notable of these is the new packaging design used for the 350 MHz DDR SDRAM

present on the circuit board. This type of packaging is called μ BGA or Micro Ball Grid Array packaging because it uses tiny solder balls instead of the more traditional solder leads. Due to the greater precision possible with the solder balls the memory can be run at greater speeds with fewer transfer errors. Another benefit of this design is the fact that since the memory is smaller and denser it can be laid out in a more space efficient manner and thus runs cooler.

Inheriting the beauty of the GeForce3

Nothing attracts gamers more than 'appearances'. And this is precisely what game developers are promising to deliver to anyone who has DirectX 8.0 support in their graphics cards. The GeForce3 was the first card to support this Microsoft API and incorporated features such as programmable vertex and pixel shaders, 3D textures and so on. The single-minded objective behind all these technologies was to give developers the freedom to program custom special effects which in turn would let them create true-to-life games. The GeForce4 takes everything a step further. It takes all the potential that the GeForce3 already had in good measure and adds a few of its own new tricks that make it a 3D card to die for.

At the heart of the GeForce4 is the